Elemental Constituents, Proximate Composition and Phytochemical Levels of Roots of *Elaeis guineensis* (Palm Tree) and *Cocos nucifera* (Coconut)

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ABSTRACT

This study aimed at ascertaining the nutritional value and medicinal properties of the roots of *Elaeis guineensis* (*E. guineensis*) and *Cocos nucifera* (*C. nucifera*). Roots of *E. guineensis* were collected from an average height tree while *C. nucifera* were from matured dwarf coconut trees growing within the University of Benin premises, Benin City, Nigeria. The samples were air-dried in the laboratory and blended into powder for the analyses. The phytochemical, proximate and elemental constituents were analyzed using standard analytical methods while the elemental composition was by flame photometry. The proximate composition of *E. guineensis* roots revealed moisture content (3.493 %), crude protein (2.64 %), crude fibre (7.00 %), crude fat (0.80 %), ash (7.33 %), and carbohydrate (70.56 %). The result for proximate composition of *C. nucifera* revealed moisture content (3.15 %), crude protein (1.75 %), crude fibre (8.83 %), lipid profile (0.20 %), ash (2.33 %), and carbohydrate (71.75 %). While the elemental analysis revealed phosphorus (0.062 %), Calcium, Ca (0.012 %), Magnesium, Mg (0.126 %), and Iron, Fe (2.110 %). This study revealed that roots of both *C. nucifera* and *E. guineensis* contain bioactive compounds that can be harnessed for medicinal use and possible inclusion in diets.

Keywords: *Elaeis guineensis, Cocos nucifera,* proximate, roots, alkaloids, saponins

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1. INTRODUCTION

Palm Trees (*Elaeis guineensis*) are evergreen, mostly tropical plant in the family *Palmae* (also known as *Arecaceae*) [1]. For identification purpose, palm trees can be classified into two sub-groups: species with leaves like fan-shaped or palmate and that which pinnate or leaves with shape like feather [2]. Palm tree has a fibrous root system or adventitious roots which indicate that they are monocots related to maize, grasses etc and they start life with one seed leaf. Palm tree roots usually grow from the stem down to the ground where they help the plant to absorb water and nutrients for their growth and development [1]. Seeds usually found on its tree have been used as paste and has been known to be effective in managing aging [5]. Toothaches have been relieved by palm tree roots. The gum from a trunk is used in treatment of diarrhoea and diseases of genito-urinary system. The date palm syrup and infusion is a good remedy for cough, fever, flu and bronchial catarrh [1]. It is known locally as "Àgbạn" in Yoruba, "akú ovinbo" in Igbo, "mosårå" in Hausa and "Ivin ebo" in Edo language respectively. It is a member of the family *Arecacae* (palm family). It is a species found in the *Cocos* genus [3]. Unlike some other plants, the palm tree has neither a tap root nor root hairs, but has a fibrous root system [4].

As loved by Indonesians, they usually refer to coconut palm as an everyday addition to life round the year. The cream extracted from the shredded kernel is used in curies and sweets and in the pacific islands for the flavouring of local dishes [5]. Husk fibres are made into

ropes and the whole husks and shells are used for fuel. The shells of coconut fruits are utilized as drinking containers and planting vessels. Sap obtained by taping the unopened spathe may be made into palm sugar or allowed to ferment into an alcoholic drink (Toddy) [5]. Dried fronds, spathes and empty bunch stalks are used for fuel [5, 6]. The coconut (*Cocos nucifera* L.) is an important fruit tree in the tropical regions, and the fruit can be made into a variety of foods and beverages [3]. They are unique species as compared to shell containing fruits as a result of their ability to hold large water quantity and in the premature state, they are referred to as jelly nuts or tender nuts and could be plucked solely for its water content [7]. Innate oil and fat are part of basic proximate composition within coconut mesophyll thus a source of energy for the plant [8]. It produces sufficient vitality during metabolism as compared to carbohydrate. In any case, the most efficient work of fat is for storage as energy source. Phytochemicals are basic plant and/or fruit composition and can be utilized as agents for management of wellbeing [9]. They decrease the chance of cancer as a result of presence of fibers, polyphenol (cancer prevention agents) and anti-inflammatory component.

The phytochemicals are created through auxiliary digestion system in generally little sums [10]. A small amount of phytochemicals are synthesized and they have a noteworthy effect on general wellbeing such as steroids, alkaloids, flavonoids, tannins, saponins, carotenoids, glycosides and phenols [10]. The aim of this study is to determine the phytochemical constituents, proximate and elemental compositions of the roots of *Elaeis guineensis* and *Cocos nucifera* plants.

2. MATERIALS AND METHODS

2.1. Sample collection and preparation: Wet samples of *Cocos nucifera* (Coconut) roots *Elaeis guineensis* (Palm tree) were gotten from a coconut tree, within the University of Benin, Ugbowo campus and taken to the Department of Crop Science, Faculty of Agricultural Sciences, University of Benin, for identification. The root samples were washed with water until dirt-free, sun-dried until constant weight and then blended to powder after which Proximate, Phytochemical and Elemental analyses were carried out on the samples.

2.2. Proximate composition: This was carried out using the method as described by the Association of Official Analytical Chemists [11].

2.3. Elemental analysis: This was carried out using atomic absorption spectrophotometric method as described by Slavin and Slavin [12]. One (1.0) g each of the samples was weighed and oven-dried into different 125 ml, Erlenmeyer flask which has been previously washed with acid and distilled water. 4 ml of perchloric acid, 25 ml HNO_3 and 2 ml concentrated H_2SO_4 were added under a fume hood. The contents were mixed and heated gently at low to medium heat on a hot plate in a fume hood. The heating was continuous until dense white fumes appeared (if any traces of carbon remain, add more concentrated $HN0_3$ and digest again to the fuming stage). Heat strongly afterwards (i.e. medium to high heat) for 5 minutes. It was allowed to cool and 40 – 50 ml distilled water was added. It was then heated for another 30 mins on the same plate at medium heat. The solution was then cooled and filtered completely with a wash bottle through a whatman no 42 filter paper (9 cm) into a 100 ml pyrex volumetric flask. It was then made up to mark with distilled water and used for the elemental analyses.

2.4. Alkaloid estimation: The quantitative estimation of alkaloids was done using the method as described by Edeoga and Eriata [13]. Empty flask was first weighed and 2.0 g each of the dried samples were weighed into the flask. The samples were dissolved in 100 ml of 10 % acetic acid and left to stand for four hours. 50 ml each of the extracts were taken into separate 250 ml conical flask and 20 ml of petroleum ether $(CH_2\ CH_2)_2O$ was added and shaken vigorously. The extracts were filtered to

remove cellular debris and then concentrated to the original volume. This was done by adding 60 ml of butanol (N-C_4H_9OH) and washed twice with l0 ml of 5 % sodium chloride (NaCl) solution. The extract was heated to dryness in a hot electrical plate[1] and the weigh was retaken.

2.5. Tannin estimation: Determination of tannin components was done by the method of Edeoga and Eriata [13]. Five (5.0) g each of the samples were weighed into separate test tubes and 10 ml 2 M HCI was added and mixed. These were shaken for 5 minutes and then transferred into volumetric flasks and made up to 50 ml. The mixtures were filtered into fresh clean tubes. 5mI each of the filtrates were introduced into separate test tubes. 3ml of 0.l M $FeCI_3$ and 3 ml of 0.008 M of potassium ferrocyanide ($K_3F_3(CN)_3$) were added to develop the colour. The absorbance was read at 505 nm wavelength within 10 minutes.

2.6. Determination of total phenol: It was evaluated using a modified colorimetric method described previously by Singleton [14]. The method involves the reduction of Folin Ciocalteau reagent by phenolic compounds, with a concomitant formation of a blue complex. 0.50 ml each of the extracts were mixed with 3.0 mL of distilled water and 0.25 ml of Folin-Ciocalteau reagent. Immediately, 0.75 ml of saturated sodium carbonate and 0.95 ml of distilled water were added. Then, the mixtures were incubated for 30 mins at 37 ºC, and the absorbance was read at 765 nm using UV-Vis spectrophotometer. The measurement was compared to a standard curve prepared with gallic acid solution. The total phenolic content was expressed as milligrams of gallic acid equivalents per gram of dry weight (mg GAE g^{-1} DW).

2.7. Saponin estimation: Determination of saponin components was done by the method of Edeoga and Eriata [13]. 2.0 g each of the samples was mixed with 100 ml of 20% methanol solution and heated overwater bath at 55 ºC for 90 minutes and allowed to cool. The extracts obtained were combined separately and concentrated to about 40 ml at 90 ^{0}C over water bath. The concentrates was transferred into separating funnels and 40 ml diethyl ether added and then shaken vigorously and allowed to stand. The ether layers were discarded while the aqueous layers were retained. 60 ml of n-butanol was added to the aqueous layers in the different separating funnels and the purification process was washed with 5% aqueous sodium chloride and evaporated to dryness in an evaporating dish in water bath and dried to constant weight in an oven.

2.8. Determination of total flavonoid: The aluminium chloride method was used for the determination of the total flavonoid content of the sample extracts [15]. Aliquots of extract solutions were taken and made up to the volume 3 ml with methanol. Then 0.1 ml $AlCl_3$ (10%), 0.1 ml Na-K tartarate and 2.8 ml distilled water were added sequentially. The test solutions were vigorously shaken. Absorbance at 415 nm was recorded after 30 minutes of incubation. A standard calibration plot was generated at 415 nm using known concentrations of quercetin. The concentrations of flavonoid in the test samples were calculated from the calibration plot and expressed as mg quercetin equivalent /g of sample.

3. RESULTS

The results of the study are presented in the tables below. The proximate composition showed the presence of carbohydrate, fibre, protein, moisture, ash, and traces of lipid. Comparatively, the palm tree root was higher in moisture, protein, ash and lipid contents while the coconut root was higher in carbohydrate and fibre contents (Table 1).

Phytochemical content showed the presence of flavonoid, phenol, alkaloid, saponin and tannin. Comparatively, the *Cocus nucifera* root presented higher flavonoid and phenol contents, while *Elaeis guineensis* root presented higher tannin content with relatively similar alkaloid and saponin content (Table 2).

On elemental composition, both root showed the presence of phosphorus, calcium, magnesium and iron. However, the *Cocus nucifera* root presented higher phosphorus and iron contents while the *Elaeis guineensis* root presented higher calcium and magnesium contents (table 3).

Table 1. The proximate composition of *Cocus nucifera* and *Elaeis guineensis* roots

	% composition per g dry weight	
Parameter	***Cocos nucifera***	***Elaeis guineensis***
Moisture contents	3.15± 0.04	3.49 ± 0.08
Crude protein	1.75± 0.05	2.64 ± 0.05
Crude fibre	8.83± 0.30	7.00 ± 0.80
Crude lipid	0.20± 0.06	0.80 ± 0.00
Ash content	2.33± 0.88	7.33 ± 0.03
Nitrogen free extracts	71.75± 1.23	70.55 ± 0.26

Values are expressed as Mean ± SEM, n=3.

Table 2. Phytochemical content of *Cocos nucifera* and *Elaeis guineensis* roots

Phytochemicals	***Cocus nucifera***	***Elaeis guineensis***
Total flavonoid content (mg quercetin /g dry material)	6.350 ± 0.15	4.4600 ± 0.400
Total phenol content (mg gallic acid equivalent /g dry material)	11.750 ± 1.45	2.6300 ± 0.210
Alkaloid (mg/g dry sample)	1.996 ± 0.002	1.940 ± 0.120
Saponin (mg/g dry sample)	3.508 ± 0.001	3.227 ± 0.030
Tannin (mg/g dry sample)	2.362 ± 0.003	4.503 ± 0.002

Values are expressed as Mean ± SEM, n=3

Table 3: The elemental composition of *Cocos nucifera* and *Elaeis guineensis* roots

	Plant sample (% dry sample)	
Elements	***Cocos nucifera***	***Elaeis guineensis***
Phosphorus	0.0620± 0.0006	0.0250 ± 0.0001
Calcium	0.0120± 0.0006	0.0277 ± 0.0026
Magnesium	0.1260± 0.0006	0.1550 ± 0.0006
Iron	2.1100± 0.0088	2.0500 ± 0.0116

Values are expressed as Mean ± SEM, n=3

4. DISCUSSION

The beneficial effects of *E. guineensis* and *C. nucifera* roots can be attributed to the cumulative effects of various biological active components present as shown in the result of this study. Previously, there are no studies conducted to explore the benefits of the roots of both plant samples. These benefits have just been studied recently as pioneered by this study. Amid the numerous amount of natural constituents present thus far identified in many plants which have exhibited safety of use, there have been known significant risk to human health at current low level of intake when used as flavouring substances [16].

Due to their natural origin, environment and genetic factors, all these can influence the chemical composition of the crude materials presented in *E.guineensis* and *C. nucifera* roots. The phytochemical component,

proximate composition and elemental analyses, as shown in this study, revealed the potentials in both human consumption and animal feeds [17].

The crude fibre contents enhances the strength provides toughness and hardness of the shell [18] which can also impose an equal response on its root although at minimal level. These also aid the ability of the moving insoluble particles from solution acting as a semi-permeable membrane which traps heavy particles in solution [18]. This can be attributed to the high carbohydrate content as nitrogen free extract of *E. guineensis* and *C. nucifera* roots as analyzed.

The presence of bioactive compounds indicates or suggests that it has a medicinal potential, this inference is drawn from the fact that most of the plants parts used in the treatment of diseases have traces of alkaloids [19]. Detected alkaloids in *Euphobia* species used as purgative. Similarly, *Momordica charanta* and *Acadirahta indica* used in the cure of malaria contain alkaloids [20, 21]. A report suggested that saponins, in low concentrations are toxic to animal mainly because they cause haemolysis of red blood cells [22]. Following the importance of flavonoids and phenols in antioxidant activity [23] reported that selected flavonoids can directly scavenge superoxides and the highly reactive oxygen-derived radicals called peroxinitrite and nitric oxide. As a result, *E. guineensis* and *C. nucifera* roots can be added as supplements in feeds and if well processed and purified can be used in medicine. The presence of phenol also in notable amount can also confer this anti-oxidant activity on it. The values for calcium and phosphorus are partially low as compared to the result of the study by [24]. High iron content was also reported by [25]. Iron being one of the most abundant metals on earth, it ranked as the most abundant metal [26] and is used in a variety of ways for example, oxidized iron chloride is included as coagulant for water treatment especially in the exclusion of heavy metals and other contaminating particles.

CONCLUSION

There is one possibility for the relevance of coconut tree and palm tree roots as a coagulant in the treatment of water and waste water and as additive to nutritional compositions. The other metals thus seen in the study, may also be useful in various forms though they could be toxic (harmful) when in large proportions or at high concentrations. It can be recommended that, preparation of decoctions from these roots should be administered with caution and regular checks.

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CONFLICT OF INTEREST

There are no conflicts of interests.

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